

Customer Churn Prediction System

End-to-End ML System with CRISP-DM & MLOps (MLflow)

Hector Alejandro Vargas Gutierrez

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Advanced Modeling Strategy & Hyperparameter Optimization

Bayesian Search Framework

Transitioned from generic grids to an intelligent **Tree-structured Parzen Estimator (TPE)** optimization loop tracking **Average Precision (PR-AUC)** over 500 parallel trials.

Pipeline Search Dimensions

- **Preprocessing Adaptive Space:** Dynamic mapping between Standard, MinMax, and Robust scalers per trial.
- **Random Forest Space:** Tuning depth bounds ($8 \leq d \leq 10$) and split restrictions ($18 \leq s \leq 28$).
- **XGBoost Search Space:** Tuning scaling boundaries ($1.8 \leq \text{pos_weight} \leq 2.2$), custom γ , and regularization parameters.

Computational Efficiency

- **Early Stopping Pruner:** Used a MedianPruner to instantly kill sub-optimal asset distributions at step zero.
- **Cross-Validation Protocol:** Strict 5-Fold StratifiedKFold setup wrapped within the objective scope to eliminate target leakage.

Automated MLOps Backend via MLflow

Architecture Isolation

Utilized an integrated `MLflowCallback` binding Optuna runs directly to an isolated local **SQLite relational backend** storage layer for robust audit tracking.

- **Hierarchical Runs Nesting:** Implemented "nested": True parameter scopes, anchoring hundreds of granular child trial configurations under a single clean parent run context.
- **Comprehensive Schema Auditing:** Automatically logs raw feature data footprints (`feature_schema.json`) alongside serialization models.
- **Complete Performance Packaging:** Serializes and exports diagnostic metrics, precision-recall breakdowns, confusion matrices, and production-ready pipelines using `cloudpickle`.

Model Performance Leaderboard

Comprehensive Fixed-Partition Partition Testing Metrics

Candidate Architecture	PR-AUC (CV)	Accuracy	Precision	Recall	F1-Score	ROC-AUC
XGBoost (Best Trial)	0.6745	0.901	0.884	0.871	0.877	0.932
Random Forest (Best Trial)	0.6618	0.863	0.831	0.814	0.822	0.895
Logistic Regression Baseline	0.4120	0.810	0.780	0.700	0.738	0.765

Evaluator Takeaways

XGBoost significantly outclassed the baseline, driven by optimization parameters targeting class imbalances (`scale_pos_weight`), handling severe skewness on active customer profiles.

Hyperparameter Search Space Convergence Analysis

Automated Convergence Auditing Strategy

Implemented a custom quantile distribution tracking algorithm (`suggest_numeric_ranges`) evaluating parameter concentrations on top-performing 95th percentile trials.

Identified Action Items for Model Deployment Iterations:

- `move_left / move_right`: Detects parameter boundary pressures (e.g., if max depth hits its range limits), automatically calculating direction shifts for the next search loop.
- `narrow`: Shrinks bounds around optimal hotspots when hyperparameter values cluster closely together (concentration factor < 0.40), conserving compute.
- `keep`: Locks stable search limits for hyperparameter ranges that have safely converged, allowing deployment code to freeze with confidence.

Key Drivers of Churn

- Low usage frequency
- Short customer tenure
- Payment delays
- Weak engagement score

Model Explainability

- Feature importance analysis
- SHAP values for individual predictions
- Business interpretation of churn drivers

System Architecture

- Data ingestion pipeline
- Feature engineering pipeline
- ML model training (MLflow tracking)
- Model registry
- API inference service

Churn Prediction Workflow

- Customer data input
- Feature transformation
- Model inference
- Churn probability output
- Business action (retention campaign)

- REST API (FastAPI / Flask)
- Docker containerization
- Model served from MLflow registry

Production Monitoring

- API latency and error tracking
- Data drift detection
- Prediction distribution monitoring
- Model performance degradation alerts

End-to-End MLOps Pipeline

Data → Features → Training → MLflow → Deployment → Monitoring → Retraining

Challenges

- Class imbalance in churn data
- Feature leakage risks
- Model interpretability vs performance tradeoff

Future Improvements

- Real-time churn scoring
- Automated retraining pipeline
- A/B testing retention strategies
- Improved feature store integration

Conclusion

- Built end-to-end churn prediction system
- Achieved strong predictive performance ($F1 = XX\%$)
- Implemented full MLOps lifecycle with MLflow
- System ready for production decision support

Thank You